

**ADMINISTRATIVE INFORMATION****FMI No: 25376****FMI Title: VCT Service Boom and PDU overheat protection update.**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date: Aug 31 st , 2006
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date: March 9 th , 2007

- ☒ **Staged FMI** **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
- ☒ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
- ☐ **Parts Ordering:** GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator
- ☐ **No Parts** are required to complete this FMI.
- ☐ **Batch with** _____
- To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
- ☐ **Non-Staged** **No Customer List is provided.** Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5160535-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25376

FMI Title: VCT Service Boom and PDU overheat protection
update.

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

The following CT Equipment is affected:

The following model numbers are tracked - 5124069 & 5123200 manufactured before January 2006.

FMI #	FMI Kit P/N	NGPDU Type	NGPDU Part No.	IB Total
25376	5260541	NGPDU-4	2326492-4	

Table 1 - Effectivity

Purpose

- The purpose of this FMI is to correct a potential failure. If the IGBT QB1 on the NGPDU control board failed, the discharge resistors RB12 through RB17 will be exposed to HVDC 700V, the result is that these resistors will get very hot and scorch the control PCB and the Lexan top cover, smoke will be emitted.
This FMI adds thermal reset circuitry to prevent the RB12 through RB17 from being exposed to 700Vdc when IGBT Q1 failed and thus avoids smoking of NGPDU.
- Check for Slipping cover label issues. This label has caused shorts across the rings.
- Update early VCT systems with newest Service Boom Design and Detector Service Kit.
- Verify Dynamic Brake fuse size on VCT systems.

Prerequisite FMIs

FMI Number	Title
	None

Related FMIs

FMI Number	Title
	None

Furnished Materials

FMI kit 5160541

Qty	Description	Part #
1	FMI Document	5160535-100
1	NGPDU-4 Overheat-protection Kit	5148601
1	Tube, Hoist arm support-VCT	5138605
2	Ring grip quick release pin	46-199678P3
1	Hoist Arm-VCT	5137423
1	Bracket-hoist extension-VCT	5137422
2	Fuse 8 AMP 250 Volts Bussmann only	46-170021P106
1	64 Slice Detector Service Kit	5149370

Tool Required

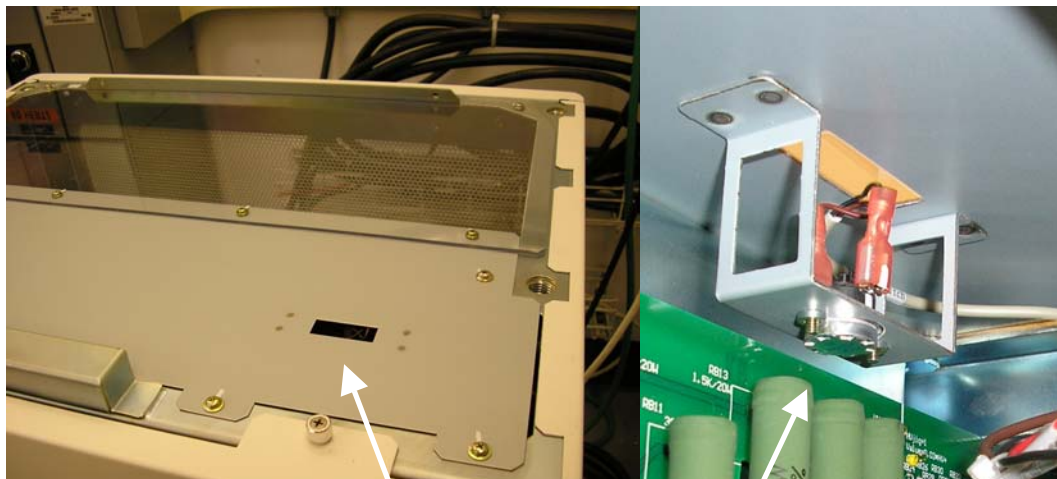
Qty	Description	SW Version	Part #
1	FMI Document	N/A	5160535-100
1	Phillips Screwdriver #2		FE Tool
1	Flat Blade Screwdriver (Small)		FE Tool
1	Long Nose Pliers		FE Tool
1	5 mm socket with ratchet wrench		FE Tool
1	Multimeter		FE Tool

Personnel Requirements

"2" hours labor and "1" hour of travel

1. Initial system inspection to verify if the system needs this FMI

Verify your system's PDU



1. If the existing PDU **does not have a metal cover with a thermal sensor**, it will need to **have this FMI completed**. Proceed to "Step 1. PDU Overheat Protection update".
2. If the existing PDU **has the metal cover with sensor**, the PDU **does not need the overheat protection**, proceed to "Step 2. Slipping cover label" procedure. Return unused Overheat protection kit 5148601 to:

HM Product Solutions, LTD.
3225 Gateway Road, Suite 250
Brookfield, WI 53045
262-754-0424

Procedure

Preparation or Before you Begin

WARNING: FOR SYSTEMS POWERED FROM A POWER DISTRIBUTION BOX THAT HAS A LOCKOUT CAPABILITY, TURN OFF AND LOCK OUT THE SYSTEM POWER AT THE MAIN INPUT BEFORE THE REPLACEMENT PROCEDURE. LOTO.

CAUTION: ELECTRICAL HAZARD ⚡

PLEASE BE ASSURED THAT YOU HAVE UNDERSTOOD ALL THE SAFETY GUIDELINES THAT YOU HAVE TO FOLLOW DURING YOUR WORK.

PLEASE TURN OFF THE MAIN POWER DISTRIBUTION SUPPLY TO CT SYSTEM AND SWITCH OFF THE CB1 BREAKER DURING YOUR FMI WORK.

CAUTION: ELECTRICAL HAZARD ⚡

WAIT SEVERAL MINUTES TO ALLOW FOR CAPACITOR DISCHARGE. VERIFY WITH A MULTIMETER BEFORE STARTING THIS PROCEDURE.

1. PDU Overheat protection update.

Follow the steps listed below:

1. Understand and follow the Safety Guidelines Manual.
2. Switch OFF the OC main power.
3. Switch OFF the breaker in the Power Distribution Box (hospital supply / A1 Panel). Perform LOTO.
4. Remove the NGPDU Top Cover.
5. Remove the Front Cover.
6. Switch OFF the CB1.
7. Remove the Top and Front Plastic covers. See Figure 1.

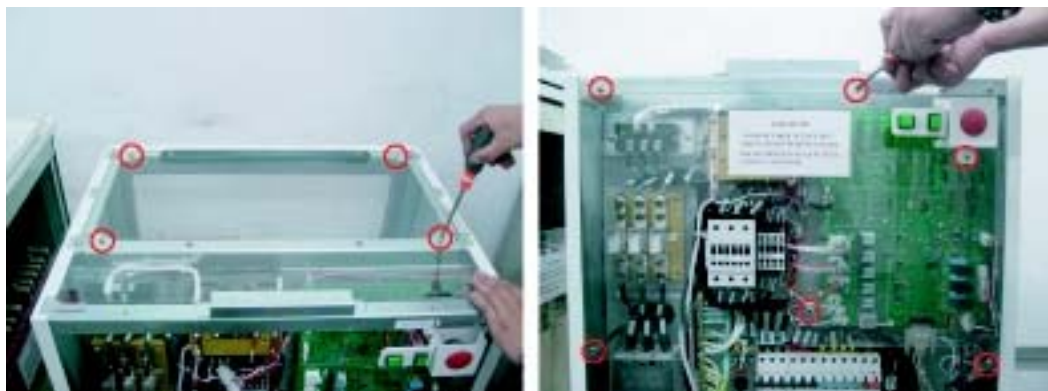


Figure 1 Overheat Protection Installation

1. Remove the old signal cable and note down cable lead connections at the Gantry switch and PDU relay.
 - a. Carefully remove the connectors of the old signal cable from the Gantry Switch, avoid causing damage to the cable or the Gantry Switch. See Figure 2 & 8-10.

Note: Please Pay Attention to Position of Connectors

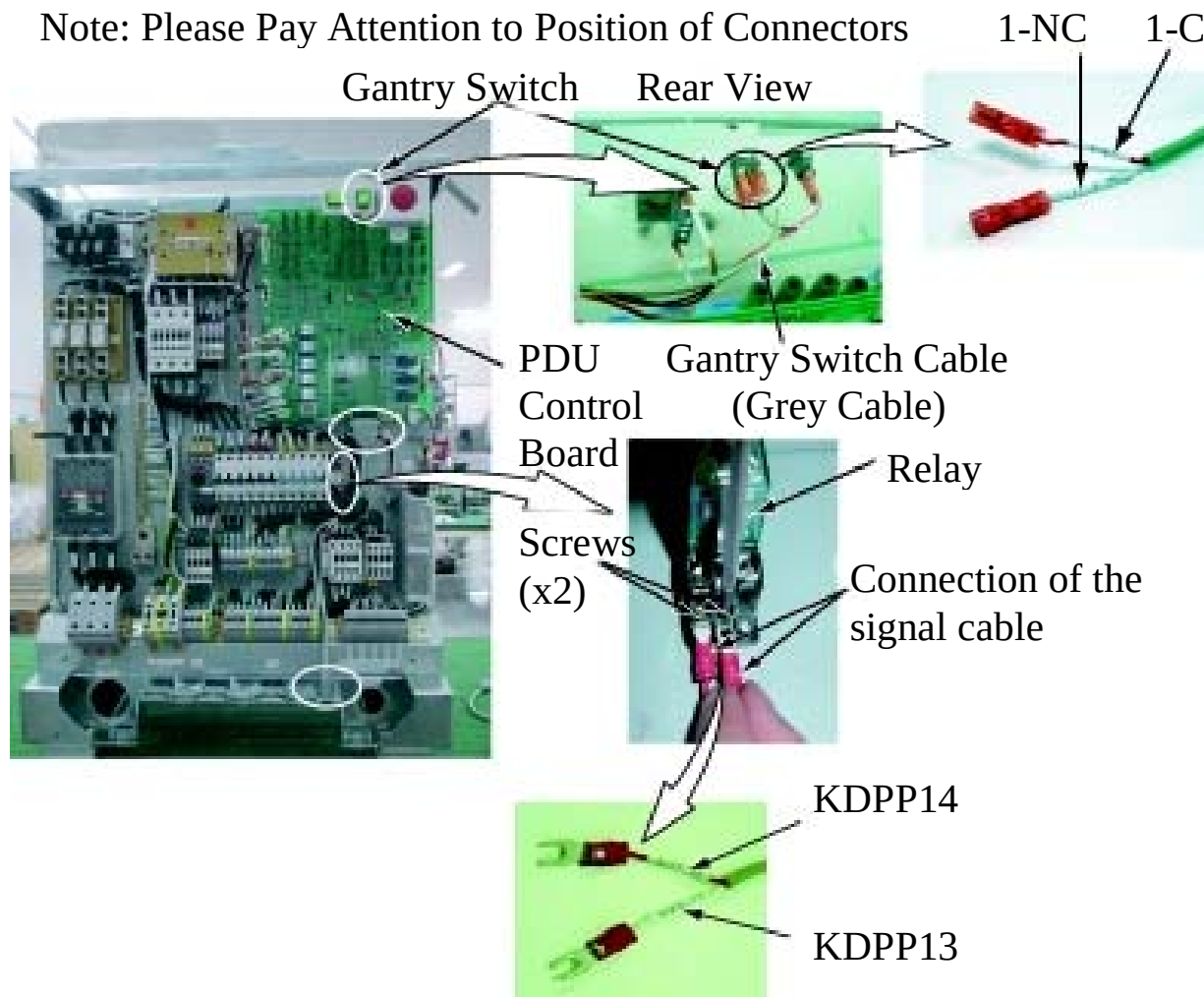


Figure 2 Remove Signal Cable

- b. Remove the connector of signal cable from the PDU Control Board (J9). See Figure 2 & 9.
 - c. Loosen the two screws that were used to secure the connectors of the signal cable (don't remove them from the relay), and then pull the two connectors out (KDPP13 & KDPP14). See Figure 2 & 9.

- d. Remove the connector of signal cable (J1) from the sheet metal by unscrewing its two screws with 5mm socket wrench. See Figure 3 & 8.

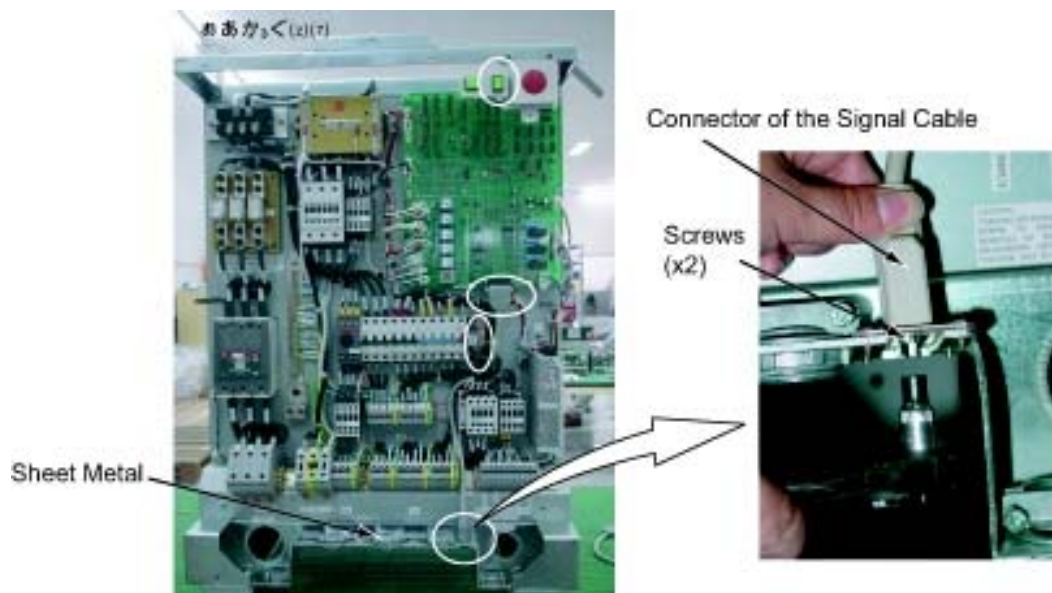


Figure 3 Remove the Signal Cable

2. Connect the Temperature Sensor to the end of the thermal switch end of the new signal cable (5147885).

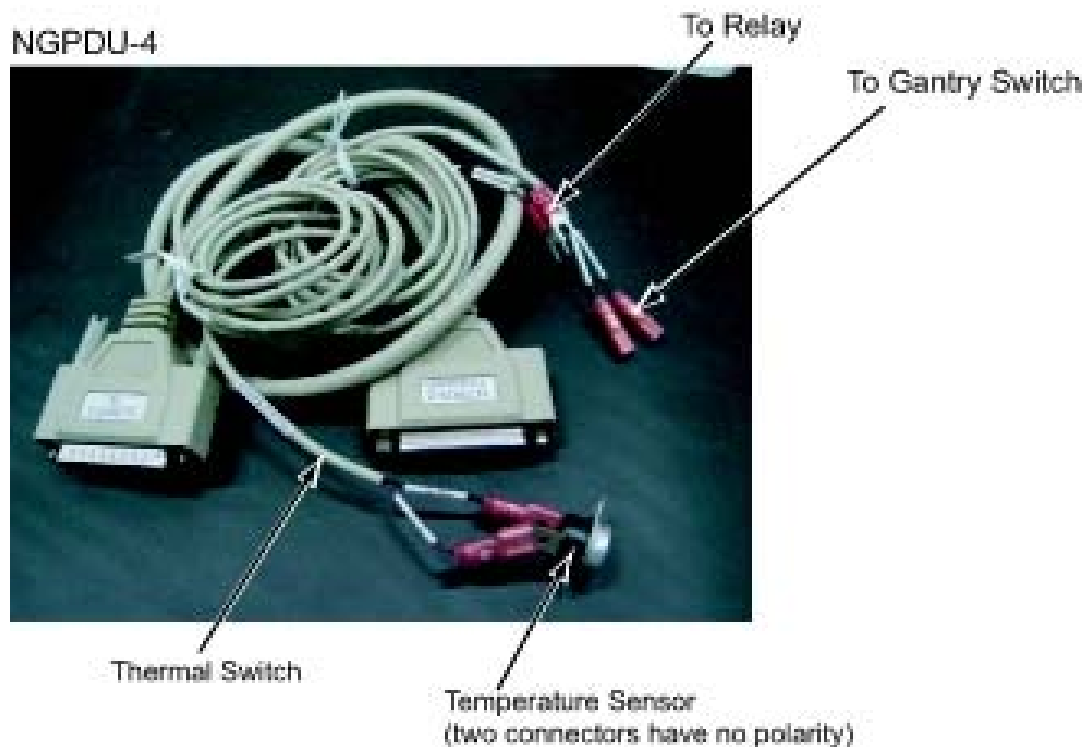


Figure 4 Install the Temperature Sensor

3. Route the Gantry Switch Cable to go behind the existing wires from the bottom up. Pass the Temperature sensor through the top of the PDU behind the front top support. See Figure 5.
4. Place the Temperature Sensor on the bracket of the sheet metal (5146322 included in FMI Kit). See Figure 5. Fasten two screws on the Temperature Sensor.

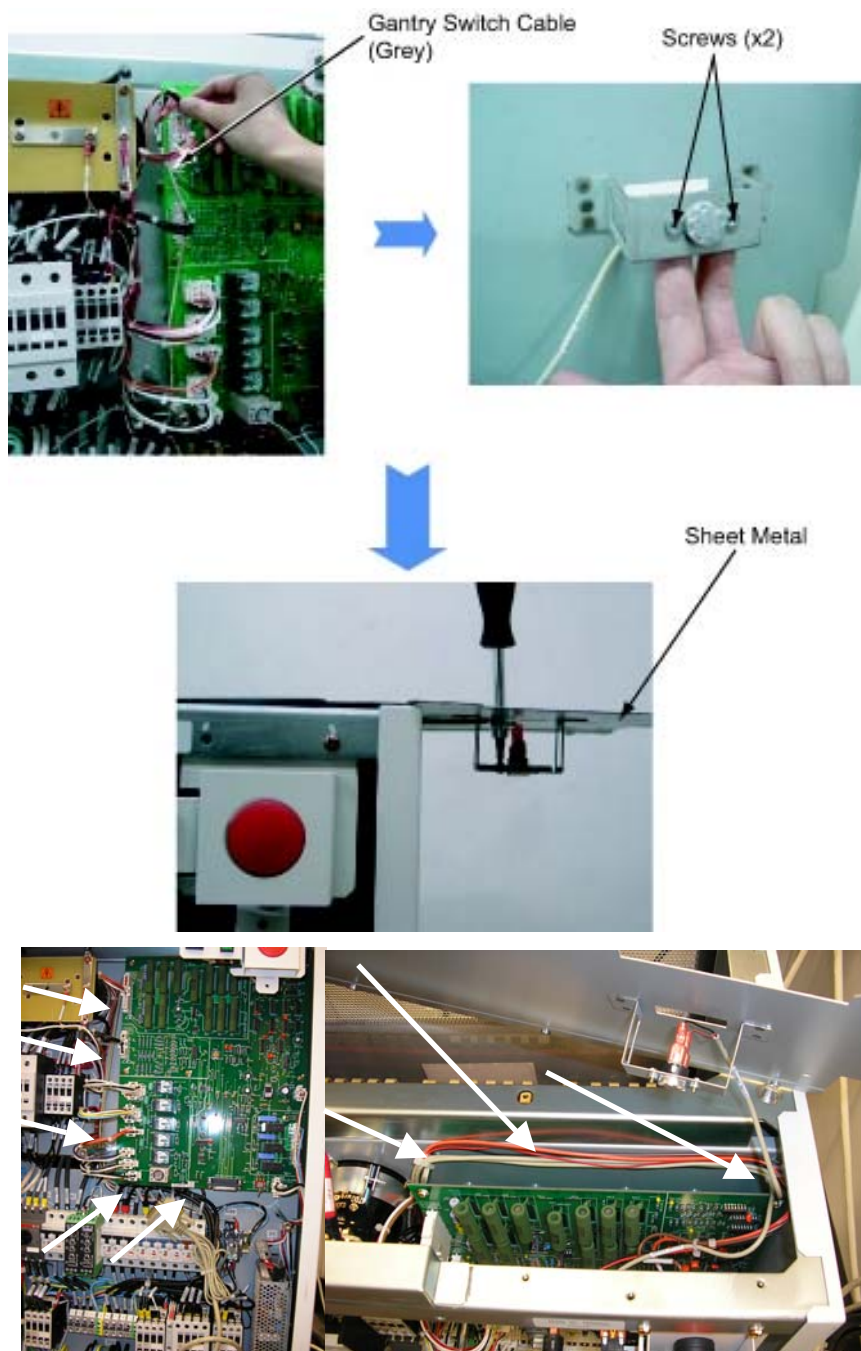


Figure 5 Route the Signal Cable and affix the Temperature Sensor on the Sheet Metal

5. Position the sheet metal (5146322) onto the top of the NGPDU-4. See Figure 6.

NOTICE: Do not install the two screws as shown in Figure 6 at this time.

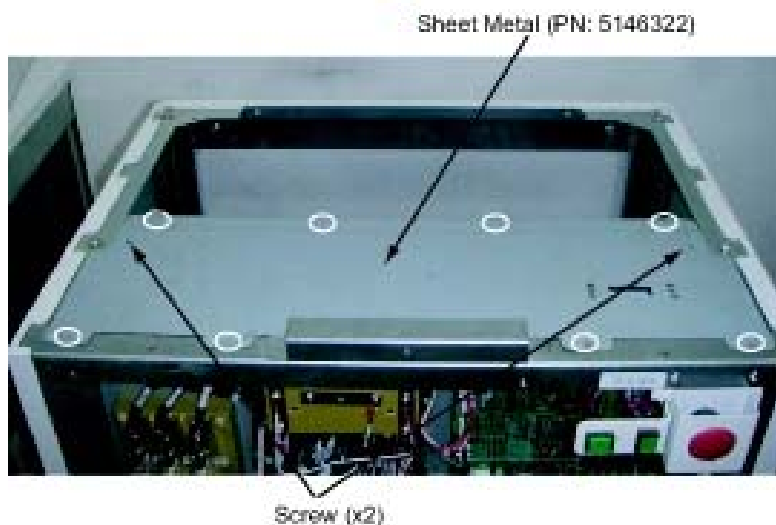
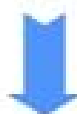


Figure 6 Install the Sheet Metal

NOTICE: For this FMI only two holes in the sheet metal are used, eight holes (white circles in above figure) are unused.

6. Connect the new signal cable to Control Board (J9), PDU Bulkhead (J1), the Gantry Switch and PDU Relay as identified in step 1. Use tie-wraps to secure the cable. See Figure 2, Figure 3, Figure 4, and Figure 7 - 10.

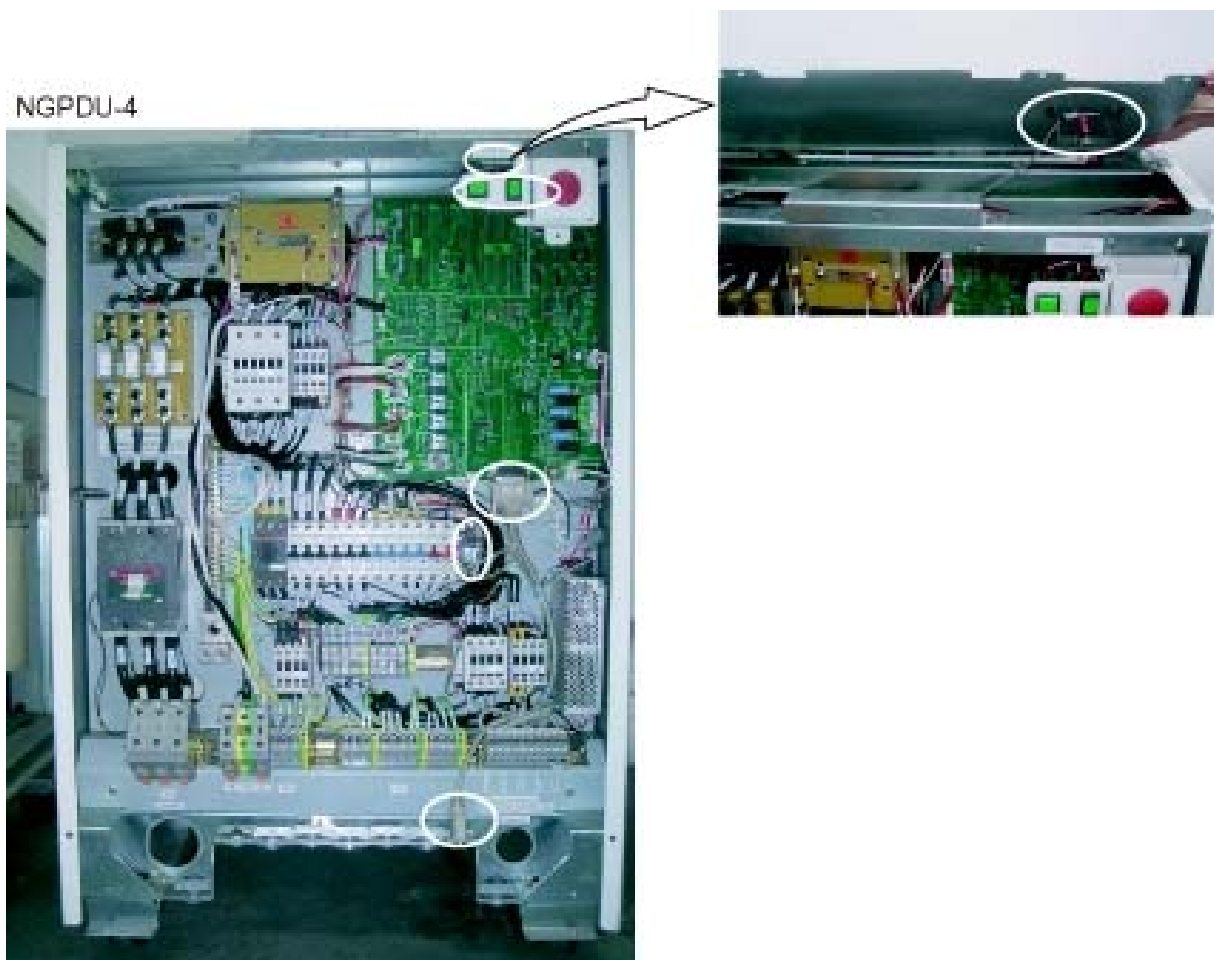


Figure 7 Install the Signal Cable

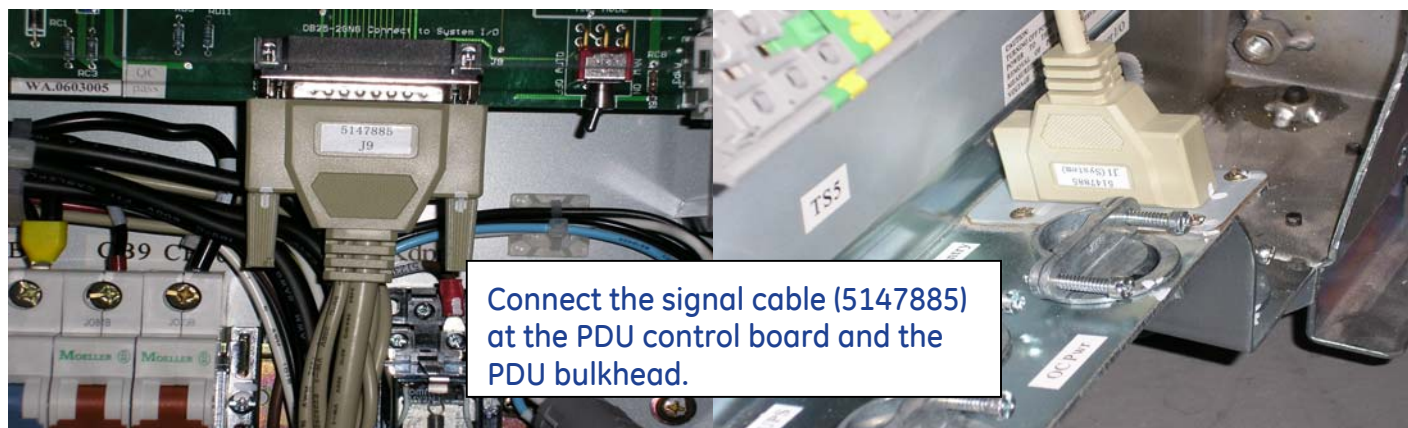


Figure 8. Signal cable connections.

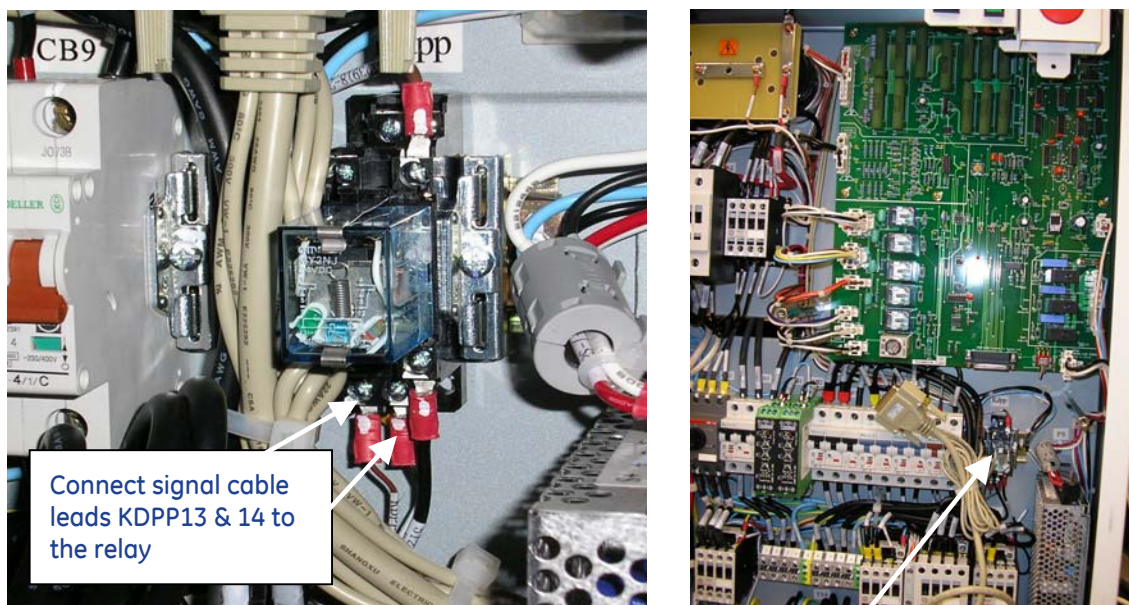


Figure 9. Relay connections. (Depending on the age of the PDU, the KDPP relay may be mounted on the right or left side of the signal cable.)



Figure 10. Gantry switch connections.

Completing overheating protection Installation

1. Install the top plastic shield (Half of the plastic shield will sit on top of the new sheet metal cover with thermal sensor).

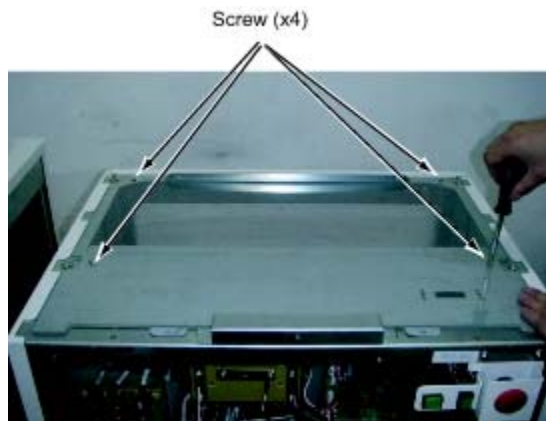


Figure 11 Install Plastic Covers

2. Install the Front Plastic Cover.
3. Switch ON the CB1 (located at the bottom left of PDU).
4. Power on the system and verify it functions normally.
5. Install the Front Cover.
6. Install the Top Cover.
7. Restore power to the system.
8. Perform test scans to verify operation.

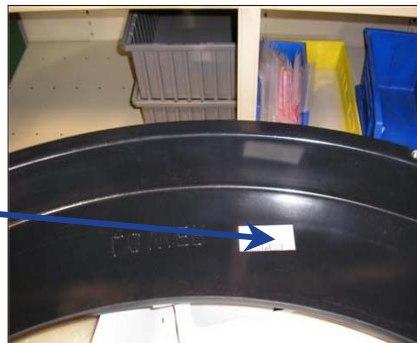
2. Slipping Cover label

- Remove power from the system and perform LOTO before servicing the Gantry.
- Remove the label from the inside of the slipping covers if it is still on the cover. Over time, this label will start to lift off the cover and can cause shorts across the rings.



Remove label from
inside of upper
Slipping cover and

Remove label from
inside of lower
Slipping cover and

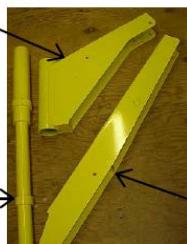


3. New Yellow Service Boom

This kit includes a new yellow service Boom. It can be used to replace or service the 64/32 slice detector. The new Universal hoist (5149069) must be used with the Yellow service boom. The hoist is available through your local service tool process.

Tube Boom

Post



Detector Boom



P/N 5149069

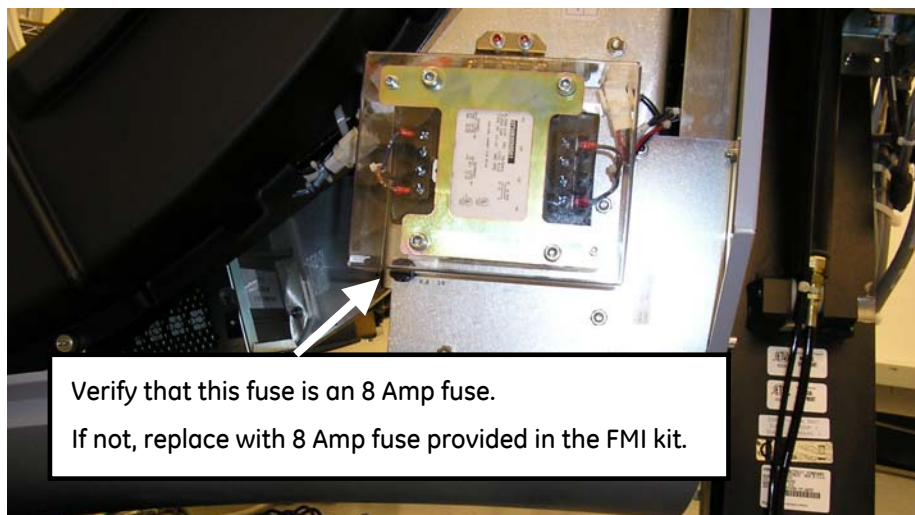


4. 64 Slice Detector Service Kit

This kit includes a 64 Slice detector service kit. No action is required at this time; please store this kit on site with the VCT system.

5. Dynamic Brake Fuse 8 Amps

- Remove power from the system and perform LOTO before servicing the Gantry.
- This kit includes an 8 AMP fuse. Please verify that the Dynamic Brake fuse F2 is an 8 Amp fuse. If the fuse is any value other than 8 Amps please replace it. If your system already has an 8 AMP fuse, please keep the one in the kit as a spare.



6. FMI Closure

Close FMI per regional procedure. Note that the FMI cannot be closed until the Product Locator cards are completed, sent in, and GIB is updated with all tracked items in this FMI.

Remove this FMI document from the customer site.